

TECHNICAL NOTE

JACOBIAN MATRICES USING THE NEWMARK DIRECT INTEGRATION SCHEME

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Abstract—The Newmark method for direct integration of nonlinear systems is considered. The differential equations of motion are written in a general form and the pertinent Jacobian matrices, which may be used as iteration matrices within the Newton-Raphson procedure, are presented. A numerical example is included.

INTRODUCTION

In a transient analysis of a nonlinear problem in applied mechanics the Newton-Raphson procedure is often used for iterations in order to find a solution for which a residual becomes zero. The pure Newton-Raphson procedure requires the establishment of the Jacobian matrix, which, by definition, contains partial derivatives of elements in the nonlinear vector-valued vector-functions involved. In structural dynamics these vector-functions form a so-called dynamical equilibrium equation having generalized coordinates as dependent variables of the independent variable, time. The dynamical equilibrium equation may be set up using the principle of virtual work, Lagrange's equations, Kane's dynamical equations, etc. which are sub-classes of a general approach known as the weighted residual method.

The iteration process may very well converge without using the Jacobian matrices. Other iteration matrices are used within the so-called fixed point iteration methods [1] having the Newton-Raphson procedure as one possible choice. The knowledge of a consistently derived Jacobian matrix is, however, a good starting point for finding such other matrices.

The studied nonlinear system of ordinary differential equations is of second order. The Newmark scheme [2], widely used in conjunction with the finite element method, will be employed for the discretization in the time domain.

In order to cover areas of much current interest such as spacecraft dynamics, vehicle dynamics, moving loads/masses, fluid-structure interaction, control of structures, etc., the dynamical equation has been given a very general form. Constraint equations [3] between generalized coordinates are, however, not considered, thus restricting the systems to have independent coordinates.

THE DYNAMICAL EQUATION

A common starting point within the finite element community is the following nonlinear dynamical equation

$$M\ddot{q} + F^{\text{int}}(\dot{q}, q) = F^{\text{ext}}(q, t), \quad (1)$$

where

M = mass matrix (assumed constant throughout the time integration)

F^{int} = vector-function of generalized internal forces

F^{ext} = vector-function of generalized external forces

$q, \dot{q}, \ddot{q}$ = vectors of generalized coordinates (displacement, velocity and acceleration, respectively)
 t = time.

It may be noted that the first term in eqn (1) is linear. The problem studied in this work is a generalization of eqn (1) into

$$F^{\text{kin}}(\ddot{q}, \dot{q}, q, t) + F^{\text{int}}(\dot{q}, q, t) = F^{\text{ext}}(\ddot{q}, \dot{q}, q, t), \quad (2)$$

where the superscript kin in the first term denotes an association with the kinetic energy.

Below is given a brief list of sources within applied mechanics that may contribute to the three nonlinear vector-functions F^{kin} , F^{int} and F^{ext} :

- F^{kin} $q, \dot{q}$: large rotations, noninertial frames of reference, Coriolis 'forces' (all typical contributions in Lagrangian dynamics)
- $\ddot{q}$: mass inertia
- F^{int} q : elastic strain
- $\dot{q}$: viscous material behaviour
- $\ddot{q}$: control
- F^{ext} q : external pressure, 'follower' forces, aeroelasticity, forces in combination with large rotations
- $\dot{q}$: drag forces in aeroelasticity and hydrodynamics (structures in fluids)
- $\ddot{q}$: 'added mass' effects in ocean engineering.

The explicit time-dependence in the vector-functions may arise from prescribed moving mass/load motions in problems such as vehicles on bridges, deploying flexible parts in spacecraft dynamics, etc.

In the following, a change of notation is made such that

$$q \equiv d, \quad \dot{q} \equiv v, \quad \ddot{q} \equiv a. \quad (3a), (3b), (3c)$$

JACOBIAN MATRICES

The Newmark time-stepping scheme [2], a one-step method, can be written:

$$v_{n+1} = v_n + \Delta t(1 - \gamma)a_n + \Delta t\gamma a_{n+1} \quad (4a)$$

$$d_{n+1} = d_n + \Delta t v_n + \frac{\Delta t^2}{2}(1 - 2\beta)a_n + \Delta t^2\beta a_{n+1}, \quad (4b)$$

where the indices n and $n+1$ refer to discrete times and $\Delta t = t_{n+1} - t_n$. The parameters γ and β may be chosen differently. A common choice is the implicit constant average acceleration method where $\gamma = 1/2$ and $\beta = 1/4$.

A residual is now defined from eqn (2) as (subsequent derivations follow, in principle, [3] and [4]):

$$\mathbf{R} \equiv \mathbf{F}^{\text{kin}} + \mathbf{F}^{\text{int}} - \mathbf{F}^{\text{ext}} \quad (5)$$

In order to find a state at time t_{n+1} that yields a zero residual, one assumes that an approximation to this state is known. The Taylor's series expansion of eqn (5) is then used as a starting point for the Newton-Raphson procedure. Since the state at $t = t_n$ is known, the knowledge of any one of the vectors $\mathbf{a}_{n+1}$, $\mathbf{v}_{n+1}$ or $\mathbf{d}_{n+1}$ uniquely describes the state using eqns (4a) and (4b). Hence, the Taylor's series expansion of the residual can be written in three forms as:

$$\mathbf{R}(\mathbf{a}_{n+1}^{i+1}) = \mathbf{R}(\mathbf{a}_{n+1}^i) + \mathbf{J}_a(\mathbf{a}_{n+1}^i)(\mathbf{a}_{n+1}^{i+1} - \mathbf{a}_{n+1}^i) + \dots \quad (6a)$$

$$\mathbf{R}(\mathbf{v}_{n+1}^{i+1}) = \mathbf{R}(\mathbf{v}_{n+1}^i) + \mathbf{J}_v(\mathbf{v}_{n+1}^i)(\mathbf{v}_{n+1}^{i+1} - \mathbf{v}_{n+1}^i) + \dots \quad (6b)$$

$$\mathbf{R}(\mathbf{d}_{n+1}^{i+1}) = \mathbf{R}(\mathbf{d}_{n+1}^i) + \mathbf{J}_d(\mathbf{d}_{n+1}^i)(\mathbf{d}_{n+1}^{i+1} - \mathbf{d}_{n+1}^i) + \dots \quad (6c)$$

which are referred to as the acceleration, velocity and displacement formulations [5] respectively. The superscript iteration-counter i denotes the current approximation and the Newton-Raphson method for the iteration procedure may be applied to eqns (6a), (6b) and (6c) in a standard way. A predictor-corrector algorithm fully described in [3] and [5] is a practical implementation scheme for this purpose and has been used in the subsequent numerical example.

The Jacobian matrix using the acceleration formulation is, by definition,

$$\begin{aligned} \mathbf{J}_a &\equiv \frac{\partial \mathbf{R}}{\partial \mathbf{a}} \\ &= \left[\frac{\partial \mathbf{F}^{\text{kin}}}{\partial \mathbf{a}} + \frac{\partial \mathbf{F}^{\text{int}}}{\partial \mathbf{a}} - \frac{\partial \mathbf{F}^{\text{ext}}}{\partial \mathbf{a}} \right] \\ &\quad + \left[\frac{\partial \mathbf{F}^{\text{kin}}}{\partial \mathbf{v}} + \frac{\partial \mathbf{F}^{\text{int}}}{\partial \mathbf{v}} - \frac{\partial \mathbf{F}^{\text{ext}}}{\partial \mathbf{v}} \right] \frac{\partial \mathbf{v}}{\partial \mathbf{a}} \\ &\quad + \left[\frac{\partial \mathbf{F}^{\text{kin}}}{\partial \mathbf{d}} + \frac{\partial \mathbf{F}^{\text{int}}}{\partial \mathbf{d}} - \frac{\partial \mathbf{F}^{\text{ext}}}{\partial \mathbf{d}} \right] \frac{\partial \mathbf{d}}{\partial \mathbf{a}} \end{aligned} \quad (7)$$

The inner derivate matrices $\partial \mathbf{v} / \partial \mathbf{a}$ and $\partial \mathbf{d} / \partial \mathbf{a}$ in this last equation become diagonal matrices and, due to the assumptions in eqns (4a) and (4b), are

$$\frac{\partial \mathbf{v}}{\partial \mathbf{a}} = \text{diag}(\Delta t \gamma), \quad \frac{\partial \mathbf{d}}{\partial \mathbf{a}} = \text{diag}(\Delta t^2 \beta). \quad (8a), (8b)$$

At this point, it is convenient to define tangent matrices. Since the dynamical equation has been given a general form the expression 'generalized tangent matrix' will be used. From eqn (7) three matrices $\mathbf{M}_t^i$, $\mathbf{C}_t^i$ and $\mathbf{K}_t^i$ are defined as (t = tangent, g = generalized)

$$\mathbf{M}_t^i \equiv \left[\frac{\partial \mathbf{F}^{\text{kin}}}{\partial \mathbf{a}} + \frac{\partial \mathbf{F}^{\text{int}}}{\partial \mathbf{a}} - \frac{\partial \mathbf{F}^{\text{ext}}}{\partial \mathbf{a}} \right] \quad (9a)$$

$$\mathbf{C}_t^i \equiv \left[\frac{\partial \mathbf{F}^{\text{kin}}}{\partial \mathbf{v}} + \frac{\partial \mathbf{F}^{\text{int}}}{\partial \mathbf{v}} - \frac{\partial \mathbf{F}^{\text{ext}}}{\partial \mathbf{v}} \right] \quad (9b)$$

$$\mathbf{K}_t^i \equiv \left[\frac{\partial \mathbf{F}^{\text{kin}}}{\partial \mathbf{d}} + \frac{\partial \mathbf{F}^{\text{int}}}{\partial \mathbf{d}} - \frac{\partial \mathbf{F}^{\text{ext}}}{\partial \mathbf{d}} \right] \quad (9c)$$

The matrices will be referred to as the generalized tangent mass matrix, the generalized tangent damping matrix and the generalized tangent stiffness matrix, respectively.

Hence, the Jacobian matrix using the acceleration formulation becomes, from eqns (7), (8a), (8b), (9a), (9b) and (9c),

$$\mathbf{J}_a = \mathbf{M}_t^i + \Delta t \gamma \mathbf{C}_t^i + \Delta t^2 \beta \mathbf{K}_t^i. \quad (10a)$$

Following the same line as above, it is readily seen that the Jacobian matrices using the velocity and displacement formulations become

$$\mathbf{J}_v = \frac{1}{\Delta t \gamma} \mathbf{M}_t^i + \mathbf{C}_t^i + \frac{\Delta t \beta}{\gamma} \mathbf{K}_t^i \quad (10b)$$

$$\mathbf{J}_d = \frac{1}{\Delta t^2 \beta} \mathbf{M}_t^i + \frac{\gamma}{\Delta t \beta} \mathbf{C}_t^i + \mathbf{K}_t^i. \quad (10c)$$

NUMERICAL EXAMPLE

In order to illustrate contributions to the generalized tangent matrices derived in the preceding section, the system in Fig. 1 is considered (see also [6, p. 245]). The system has two degrees-of-freedom, x and θ , and consists of a horizontally moving mass M connected to a rigid wall via a linear spring with stiffness k . A shaft, the axis of which is mounted normal to the plane of Fig. 1, supports without friction a massless rod that carries a concentrated mass m at the tip. A gravitational field with constant acceleration g is assumed to exist. An explicitly time-dependent force F , constant in direction, acts on the mass m .

The governing equations of motion for the system may be derived using Lagrange's equations. One finds [6]

$$(M + m)\ddot{x} + mL(\ddot{\theta} \cos \theta - \dot{\theta}^2 \sin \theta) + kx = F \quad (11a)$$

$$mL\ddot{x} \cos \theta + mL^2\ddot{\theta} = FL \cos \theta - mgL \sin \theta. \quad (11b)$$

The associated vector-functions described above are

$$\mathbf{F}^{\text{kin}} = \begin{bmatrix} (M + m)\ddot{x} + mL(\ddot{\theta} \cos \theta - \dot{\theta}^2 \sin \theta) \\ mL\ddot{x} \cos \theta + mL^2\ddot{\theta} \end{bmatrix} \quad (12a)$$

$$\mathbf{F}^{\text{int}} = \begin{bmatrix} kx \\ 0 \end{bmatrix} \quad (12b)$$

$$\mathbf{F}^{\text{ext}} = \begin{bmatrix} F \\ FL \cos \theta - mgL \sin \theta \end{bmatrix} \quad (12c)$$

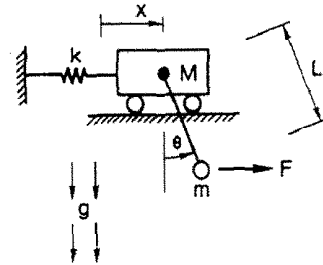


Fig. 1. Two-degree-of-freedom system studied in numerical example. Numerical data are $m = M = 0.1$ kg, $k = 10$ N/m, $L = 0.1$ m and $g = 9.81$ m/sec².

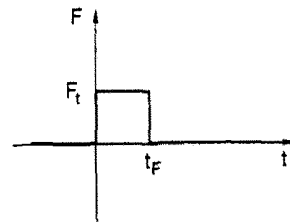


Fig. 2. Time history of applied load shown in Fig. 1. Numerical data are $F_t = 1$ N and $t_F = 0.1$ sec.

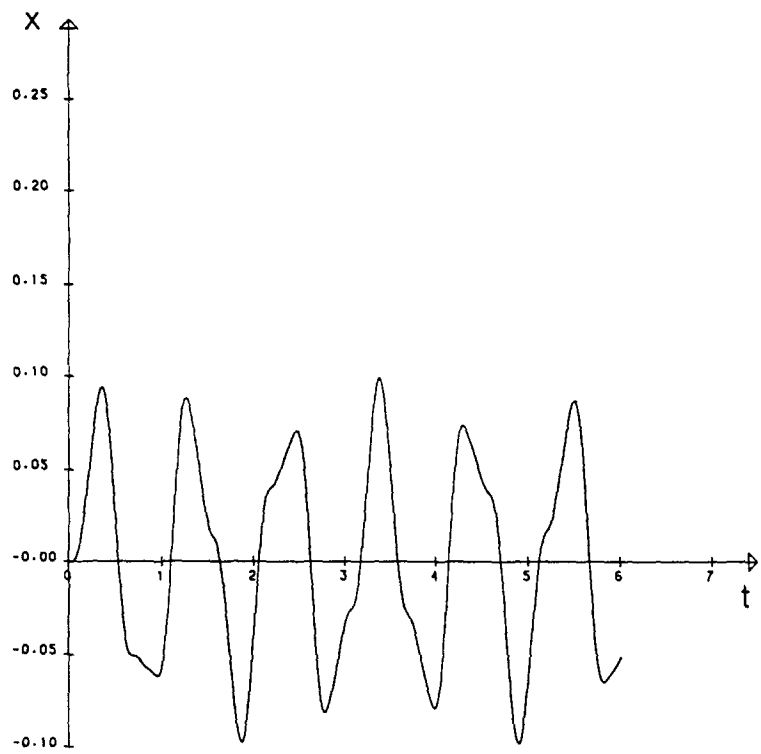


Fig. 3(a). Computed translation x (m) vs time for system in Fig. 1 with applied load shown in Fig. 2.

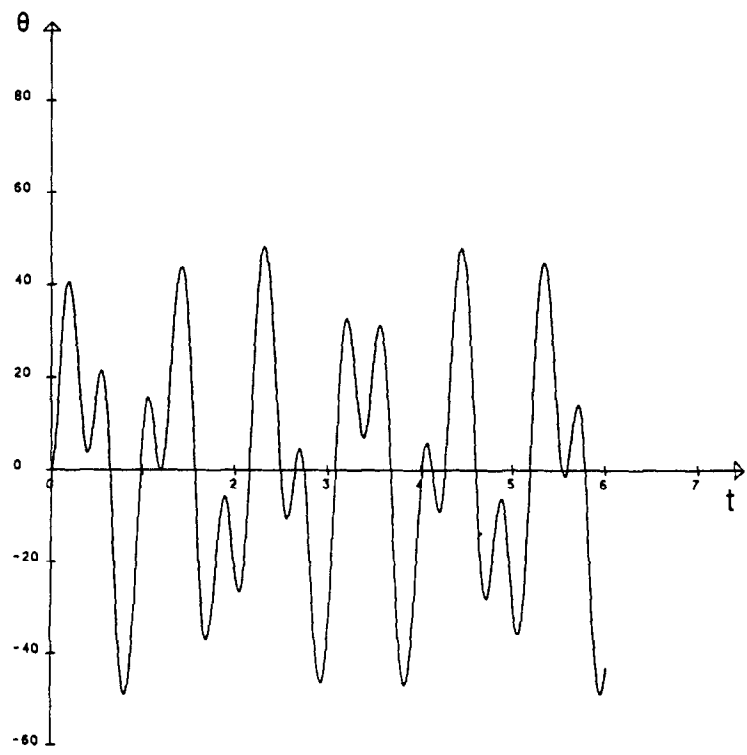


Fig. 3(b). Computed rotation θ (°) vs time for system in Fig. 1 with applied load shown in Fig. 2.

The contributions to the tangent matrices defined in eqns (9a), (9b) and (9c) thus become (with $\mathbf{d} = [x \ \theta]^T$, $\mathbf{v} = [\dot{x} \ \dot{\theta}]^T$, etc.)

$$\frac{\partial \mathbf{F}^{\text{kin}}}{\partial \mathbf{a}} = \begin{bmatrix} M + m & mL \cos \theta \\ mL \cos \theta & mL^2 \end{bmatrix} \quad (13a)$$

$$\frac{\partial \mathbf{F}^{\text{int}}}{\partial \mathbf{a}} = 0, \quad \frac{\partial \mathbf{F}^{\text{ext}}}{\partial \mathbf{a}} = 0 \quad (13b), (13c)$$

$$\frac{\partial \mathbf{F}^{\text{kin}}}{\partial \mathbf{v}} = \begin{bmatrix} 0 & -2mL\dot{\theta} \sin \theta \\ 0 & 0 \end{bmatrix} \quad (13d)$$

$$\frac{\partial \mathbf{F}^{\text{int}}}{\partial \mathbf{v}} = 0, \quad \frac{\partial \mathbf{F}^{\text{ext}}}{\partial \mathbf{v}} = 0 \quad (13e), (13f)$$

$$\frac{\partial \mathbf{F}^{\text{kin}}}{\partial \mathbf{d}} = \begin{bmatrix} 0 & -mL(\ddot{\theta} \sin \theta + mL\dot{\theta}^2 \cos \theta) \\ 0 & -mL\ddot{x} \sin \theta \end{bmatrix} \quad (13g)$$

$$\frac{\partial \mathbf{F}^{\text{int}}}{\partial \mathbf{d}} = \begin{bmatrix} k & 0 \\ 0 & 0 \end{bmatrix} \quad (13h)$$

$$\frac{\partial \mathbf{F}^{\text{ext}}}{\partial \mathbf{d}} = \begin{bmatrix} 0 & 0 \\ 0 & -FL \sin \theta - mgL \cos \theta \end{bmatrix} \quad (13i)$$

It may be observed that the resulting generalized tangent matrices are nonsymmetric (some contributions are symmetric).

The starting acceleration vector at time $t = t_0$ is solved from the system of equations

$$\begin{bmatrix} M + m & mL \cos \theta_0 \\ mL \cos \theta_0 & mL^2 \end{bmatrix} \begin{bmatrix} \ddot{x}_0 \\ \ddot{\theta}_0 \end{bmatrix} = \begin{bmatrix} F(t_0) - kx_0 + mL\dot{\theta}_0^2 \sin \theta_0 \\ F(t_0)L \cos \theta_0 - mgL \sin \theta_0 \end{bmatrix} \quad (14)$$

with specified initial conditions for x_0 , θ_0 and $\dot{x}_0$, $\dot{\theta}_0$.

The load F is assumed to have a time history as defined in Fig. 2. In Figs (3a) and (3b) the computed translation

$x(m)$ and the rotation $\theta(^{\circ})$ are plotted vs time using a constant time-step $\Delta t = 1.25 \times 10^{-2}$ sec throughout the integration. The initial conditions applied are $x_0 = \dot{x}_0 = \theta_0 = \dot{\theta}_0 = 0$. The time-integration parameters used are $\gamma = 1/2$ and $\beta = 1/4$.

CONCLUSIONS

Jacobian matrices for general second-order systems have been derived using the Newmark time-discretization scheme. Future research work of interest includes the selection of time-steps adaptively, i.e. an automatic time-stepping algorithm [7].

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